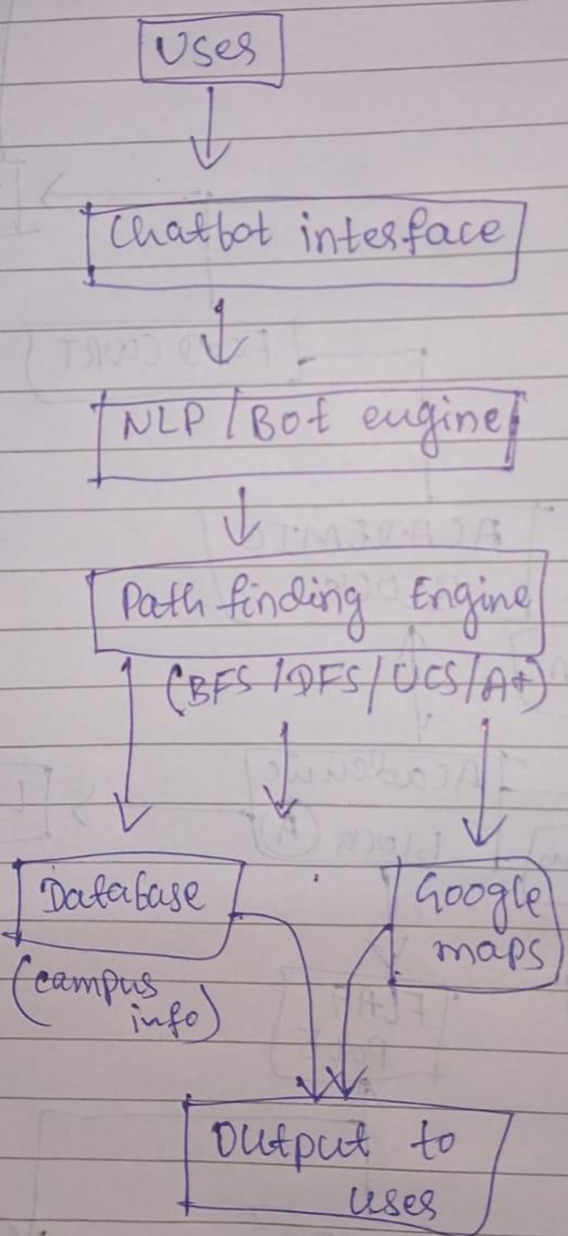
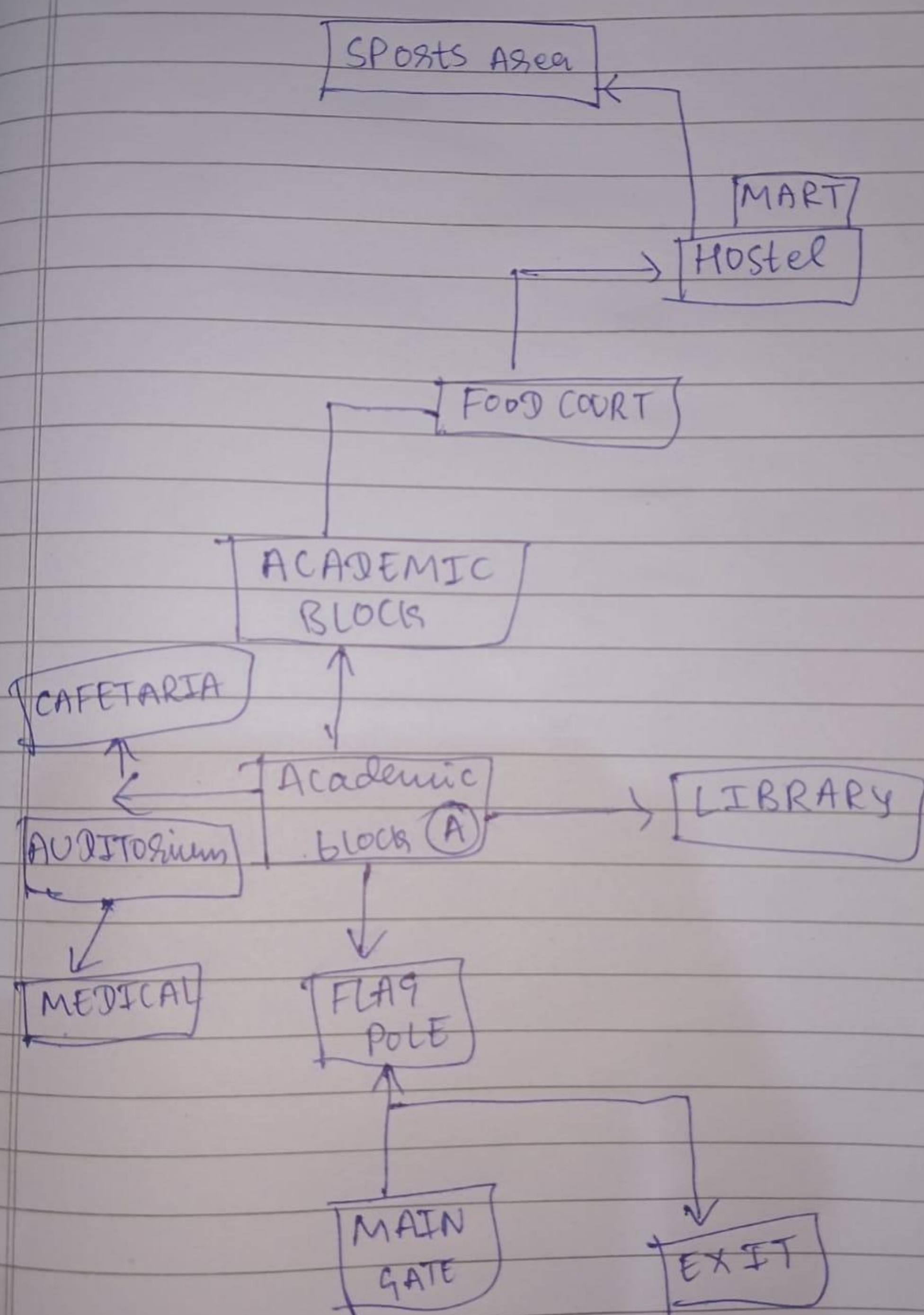
Date :
Page No.:

System Architecture Diagram



BLOCK DIAGRAM



1. System Architecture Diagram

The architecture is designed with a modular flow:
User → Chatbot UI → NLP & Bot Engine → Navigation/Pathfinding → Database
Explanation:
User asks a question.
Chatbot UI receives it.
NLP engine interprets meaning.
Pathfinding engine (BFS, DFS, UCS, A*) solves navigation queries.
Database provides FAQs, staff info, or building details.

2. Modules & Features

User Interface (UI): User interacts via chatbot (console/text).
NLP & Bot Engine: Identifies query type (pathfinding, FAQ, staff info).
Navigation/Pathfinding: Algorithms BFS, DFS, UCS, A* provide routes.
Database: Stores building details, staff info, FAQs (can start with Python dictionaries).

3. Database Schema Design

Locations Table: location_id, name, description, timings.
Paths Table: path_id, source, destination, distance.
Staff Table: staff_id, name, department, office_location, contact.
FAQs Table: faq_id, question, answer.
Even if not fully implemented, showing schema proves system planning.

4. Methodology

Modeled the campus as a graph (nodes = places, edges = paths).
Implemented BFS, DFS, UCS, A* for navigation.
Designed modular architecture (UI, NLP, Pathfinding, DB).
Created database schema for structured storage.
Tested with sample queries like "Find path from Library to Hostel".

5. Sample Pathfinding Example

Query: Find path from Library to Hostel.
Path found: Library → Student Center → Hostel
Distances:

Library → Student Center

Student Center → Hostel

This shows BFS, DFS, UCS, and A* work for campus navigation.